

Tiziana A. Gelmi Candusso, PhD

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RESEARCH INTERESTS

Wildlife Conservation, Landscape Ecology; Human-Wildlife Coexistence; Urban Ecology; Species Interaction; Forest ecosystem; Ecosystem Services; Species distribution; Habitat Conservation

ACADEMIC APPOINTMENTS

Postdoctoral Fellow , Urban Wildlife Institute, Chicago IL, USA	2024–current
Research focus: Human-wildlife coexistence in urban areas (advisors: Prof. Dr. Seth Magle; Dr. Maureen Murray, Dr. Mason Fidino)	
Postdoctoral Fellow , University of Toronto, ON, Canada	2019–2024
Research focus: Wildlife ecology in urban areas (advisor: Prof. Marie-Josée Fortin)	

EDUCATION

Ph.D. in Biodiversity and Ecology , Georg-August Göttingen University	2019
developed at Behavioral Ecology department at the German Primate Center Committee: Prof. Eckhard W. Heymann, Prof. Pierre-Michel Forget, Prof. Oliver Schülke	
M.Sc. in Environmental Biology , University of Trieste, Italy	2012
B.Sc. in Biology , University of Padua, Italy	2010

Additional Education

Science Policy and Advocacy Certificate program – University of California Irvine	2023
Forecasting for Decision-making. Short school – CEFI	2023
AI Computer vision methods for Ecology Summer school – Caltech	2022
Agent-based modeling with NetLogo. Summer school – Göttingen University	2017
Spatially-Explicit Modeling of Ecological Systems in R. Summer course – Würzburg U	2017
GIS for ecology, and Species distribution modeling – Marburg University	2016
Modelling of Biogeochemical cycles. Summer school – Christian-Albrecht-Universität	2012
Taxonomy and Physiology of marine fauna. Summer school – Sorbonne Université	2011

TEACHING

Courses

Full Course: EEB491 Seminar in Biodiversity and Conservation Biology	2026
Undergraduate level. Ecology and Evolutionary Biology. University of Toronto	
Full course: Animal behavior and Conservation in Urban Landscapes	2025
Undergraduate level. Environmental Science. DePaul University	
Invited lecture: Camera traps for Urban Ecology	2024

Course: Forest Conservation Biology. Graduate level. Daniels' faculty. University of Toronto

Invited lecture: Landscape Ecology for Conservation 2023

Course: Spatial Ecology. Graduate level. University of Toronto

Invited lecture: Animal Seed dispersal 2019

Course: 408. Primate ecology. Graduate level. Göttingen University

Course development

ENV 390 and 490 : Animal behavior and Conservation in Urban Landscapes 2024

Undergraduate level. Environmental Science. DePaul University

EEB125H1: Introductory Computation and Data Science for the Life and Physical Sciences. Ecology and Evolutionary Biology, University of Toronto 2021-2022

Workshops

Using OpenStreetMaps for LULC maps – IUWC Conference Workshop Jun 2025

Spatial Visualization in R– UWIN workshop series Jun 2024

Spatial Analysis in R– Toronto Region Conservation Authority May 2024

Landscape connectivity in Julia – Omnidirectional connectivity with Omniscape Nov 2021, Sep 2022

Seminars

Insights into our local nocturnal wildlife Jan 2025

Meeting of The Brodie Club, planned for January 21, 2025

Considering Wildlife coexistence in Urban Planning Nov 2024

Ontario Professional Planners Institute: Webinar Wildlife & Urban Planning

Maintaining landscape connectivity for reaching Biodiversity targets Aug 2023

Toronto Regional Conservation Authority

The interaction between Wildlife, Landscape and People Jun 2024

School of Cities, **University of Toronto**

The Ecology of Urban mammals in Toronto Nov 2022

UTM Biology Seminar Series. **University of Toronto Mississauga**

Mentoring

Final year undergraduate research projects 2022-2023

BIO98 Janice Hau, Juan de los Rios. Introduction to ArcGIS Pro, Julia, R.

Connectivity analysis and urban mammalian biodiversity

BIO98 - Jelany Duali. Urban Wildlife Ecology

Summer undergraduate research projects 2022

NSERC Germain Collinge Menard, BIO98 Jeffrey Chen, BIO98 Devni

Kumarasinghe

Introduction to R and spatial-occupancy analysis. Predator-Prey interactions

Master thesis 2021

Nikole Dimitrov. Agent-based Modelling using NetLogo

Undergraduate thesis

2021

Henry James. Landscape Connectivity assessment using Julia

Field Work – Data collection methods

2020–2021

21 undergraduates, training on field biosafety, on-site sampling, and camera trap data processing

PUBLICATIONS**Journal papers**

- 2025 **Gelmi-Candusso TA**, Chin ATM, Ruppert JR, Fortin MJ. (2025) Urban planning for wildlife connectivity, a multispecies assessment of urban sprawl and SLOSS renaturalization strategies. **Journal of Applied Ecology**
- Murray MH, Surasinghe T, **Gelmi-Candusso TA**, Sander HA, Fidino M, Adalsteinsson S, Rivera K, Flores A, Lilly M, Diuk-Wasser MA, Magle A. (2025) Multi-city research networks are needed to address global One Health challenges. **Bioscience**
- Murray MH, **Gelmi-Candusso TA**, Short Gianotti AG, Morzillo AT, Young JK, Larson KL, Fidino M, Magle S, Riley SPD, Sikich JA, Schell CJ and Wilkinson CE (2025) Health as an outcome and driver of human–wildlife interactions. **Bioscience**
- Devarajan, K., Fidino, M., Farris, Z. J., Adalsteinsson, S. A., (...) **Gelmi-Candusso, T. A.**, (...) A., Zellmer, A. J., and Gerber, B. D. (2025) When the wild things are: Defining mammalian diel activity and plasticity. **Science Advances**
- 2024 **Gelmi-Candusso TA**, Chin ATM, Thompson CA, McLaren AAD, Wheeldon TJ, Patterson BR, Fortin MJ. (2024) Dynamic connectivity assessment for a terrestrial predator in a metropolitan region. **Frontiers in Ecology and the Environment**. <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/fee.2633>
- Fidino, M., Sander, HA., Lewis, JS., Lehrer, EW., Rivera, K., Murray, MH., Adams, HC., Kase, A., Flores, A., Stankowich, T., Schell, CJ., Salsbury, CM., Rohnke, AT., Jordan, MJ., Green, AM., Gramza, AR., Zellmer, AJ., Williamson, J., Surasinghe, TD., Storm, H., Sparks, KL., Ryan, TJ., Remine, KR., Pendergast, ME., Mullen, K., Minier, DE., Middaugh, CR., Mertl, AL., McClung, MR., Long, RA., Larson, RN., Kohl, MT., Harris, LR., Hall, CT., Haight, JD., Drake, D., Davidge, AM., Cheek, AO., Bloch, CP., Biro, EG., Anthonysamy, WJB., Angstmann, JL., Allen, ML., Adalsteinsson, SA., Gianotti, AGS., LaMontagne, JM., **Gelmi-Candusso, TA.** and Magle, S.B. (2024) Gentrification drives patterns of alpha and beta diversity in cities. **Proceedings of the National Academy of Sciences** <https://www.pnas.org/doi/10.1073/pnas.2318596121>
- Gelmi-Candusso, T.**, Rodriguez, P., Fidino, M., Rivera, K., Lehrer, E., Magle, S. and Fortin, M.-J. (2024), Leveraging Open-Source Geographic Databases to Enhance the Representation of Landscape Heterogeneity in Ecological Models. **Ecology and Evolution**, 14: e70402. <https://doi.org/10.1002/ece3.70402>
- 2023 **Gelmi-Candusso**, Wheeldon TJ, Patterson B, Fortin MJ. (2023) The effect of urbanization and behavioral factors on coyote net displacement and its implications

- for seed dispersal. **Urban Ecosystems**. <https://doi.org/10.1007/s11252-023-01460-7>
- Gelmi-Candusso TA**, Rodriguez P, Fortin M-J. (2023) Unveiling Urban Complexity: Enhancing Fine-Scale Landscape Heterogeneity through OpenStreetMap Integration. **Biorxiv**. <https://doi.org/10.1101/2023.10.31.564785>
- Gelmi-Candusso TA**, Brimacombe C, Collinge Menard G., Fortin MJ. (2023) Generating predator-prey networks from camera traps. **Food webs** 37. <https://doi.org/10.1016/j.fooweb.2023.e00305>
- 2019 **Gelmi-Candusso TA**, Hämäläinen, A. (2019) Seeds and the city: the interdependence of zoochory and ecosystem dynamics in urban environments. **Frontiers in Ecology and Evolution**. <https://doi.org/10.3389/fevo.2019.00041>
- Gelmi-Candusso TA**, Bialozyt, R., Slana. D., Zarate-Gomez R., Heymann, E. W., and Heer, K. (2019). Estimating seed dispersal distance: a comparison of methods using animal movement and plant genetic data on two primate-dispersed Neotropical plant species. **Ecology and Evolution**. <https://doi.org/10.1002/ece3.5422>
- Gelmi-Candusso TA** (2019) Frugivore behavior and plant spatial genetics. **Doctoral dissertation**. Georg-August-Universität, Göttingen, Germany. <http://hdl.handle.net/11858/00-1735-0000-002E-E5C8-C>
- 2017 **Gelmi-Candusso TA**, Heymann, EW., and Heer, K. (2017). Effects of zoochory on the spatial genetic structure of plant populations. **Molecular Ecology**. <https://doi:10.1111/mec.14351>.

PRESENTATIONS

Invited Talks

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| June 2025 | “Understanding the distribution of sarcoptic mange across cities: Habitat loss, human pressure and climate change, a mange trifecta.” International Urban Wildlife Conference, Atlanta GA. |
| June 2024 | “Movement corridors for urban wildlife: Pivotal infrastructure for carnivores” North American Congress for Conservation Society (NACCB), Vancouver, Canada |
| June 2023 | “The impact of landscape connectivity on urban biodiversity” –International Urban Wildlife Conference, Washington DC. |
| Sep 2022 | “Training a machine learning object classifier for the urban ecosystem” – Camtrap Ecology meets AI Workshop |
| Aug 2021 | “Dynamic connectivity for coyotes in the Toronto region” The International Society for Ecological Modelling Global Conference. <i>Cancelled due to Travel Restrictions</i> |
| May 2021 | “Landscape connectivity for coyotes in the Toronto region” Atwood EEB, University of Toronto |
| May 2021 | “Key urban features affecting mammalian mesopredator movement across cities” International Urban Wildlife Conference, Portland, Oregon |

Conference Talks

Aug 2021	“Coyotes as seed dispersers in urbanized landscapes”, CSEE–SCEE, Vancouver
May 2020	“Landscape connectivity for coyotes in the Toronto region” IALE - Toronto
Mar 2020	“Urban Seed dispersal” Frugivore Seed Dispersal Conference, India
Feb 2017	“How does territoriality behavior of seed dispersers affect spatial genetics of dispersed plants?” European conference of Tropical Ecology - Brussels, Belgium
Sep 2016	“The influence of animal behavior on plant population genetics” Annual Meeting of the Ecological Society of Germany, Austria, and Switzerland - Marburg, Germany

RESEARCH ASSISTANTSHIP

Research Assistant , Max Planck for multidisciplinary sciences	2012-2013
Master student , Helmholtz Institut GEOMAR	2011-2012
Research intern , <i>Marine mammal behavior lab</i> , University of Victoria	Jul 2011
Research intern , <i>Ecotoxicology lab</i> , Università di Trieste	Ago 2010
Research intern , <i>Taxonomy lab</i> , Area Marina Protetta Miramare Trieste	Sep 2010

PROFESSIONAL SERVICE

Academic networks	Urban wildlife information network UWIN (lead for U of Toronto) The Urban Wildlife Working Group TWS WILDLABS. The conservation technology network British Ecological Society BES Canadian Society for Ecology and Evolution CSEE The Ecological Society of Germany, Austria and Switzerland Gfö
Committee member	Technology committee, One Health health committee, and Administrative committee at Urban Wildlife Information Network (UWIN)
Peer review	Journal of Applied Ecology, Ecology and Evolution, Food webs, Scientific reports, Movement Ecology, Conservation Biology, Urban Ecosystems
Grant review panel	EEB PhD research grant, University of Toronto, 2021
Award jury panel	EEB Undergraduate Research Fair, University of Toronto, 2022
Symposiums organized	Urban animal movement Symposium, International Urban Wildlife Conference, May 2021 Urban Seed Dispersal Symposium Frugivores and Seed dispersal conference, Mar 2020
Conference organized	Woman’s Career Network conference Göttingen, Germany, 2018
Community outreach	Introduction to camera traps and urban ecology for School-aged students 2021, 2023 Science communication posts on plant spatial genetics for the Frugivore and Seed dispersal (FSD) webpage for General public - 2019

In-situ introduction to the ecology of marine protected areas for young school-aged children – 2010

Media presence CBC Metro News, Toronto Star

AWARDS AND GRANTS

NSF – National Science Foundation Postdoctoral Fellowship (24m, 150,000 USD)	Sep 2024
MITACS Postdoctoral Internship (8m, 35,000 CAD)	Nov 2023
School of Cities, University of Toronto Postdoctoral fellowship (1y, 30,000CAD)	May 2023
DFG - German Research Foundation, Postdoctoral fellowship (3y6m, 271,800 CAD)	Nov 2019
Georg-August- Göttingen university grant (6m, 16,610CAD)	Jun 2017
European Commission Erasmus placement (8m, 6,040CAD)	Jan 2012
University of Trieste Scholarship (2y, 4,530CAD)	Oct 2010
University of Padova Excellence scholarship (3y, 27,180CAD)	Oct 2006

LANGUAGE AND SKILLS

Language	English (fluent), Spanish (fluent), Italian (fluent), German (advanced), French (advanced)
Programming	R, Python, Julia, SQL, GIS, NetLogo, Java, HTML, Machine Learning for computer vision
Software	JAGS, R studio, VSC, ArcGIS Pro, PostgreSQL, Compute Canada, Omniscape
Data types	Remote sensing, GPS devices, Camera Traps, Microsatellites, Rodent traps, Surveys